Measuring Muscle Fatigue With Electromyography

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Abstract

Muscle fatigue can be difficult to measure due to the wave signal. Due to being generated by an EMG-sensor, it is not always accurate and therefore can be challenging to read. As such, obtaining correct data from the electric signal generated by the muscle is crucial for the determination of muscle fatigue. This project aims to collect as much data as possible and develop a pattern recognition system that can determine if the muscle is exhibiting signs of fatigue. The expected result is to identify a measurable pattern in the EMG signal that can distinguish between a fatigued and non fatigued muscle.

Introduction

Surface electromyography (sEMG) can be used to evaluate muscle fatigue by measuring the electrical activity of the muscle. However, the signal is affected by factors such as noise, sensor properties and electrode placement. Therefore making reliable signal acquisition a priority before fatigue can be analyzed [1], [2]. Previous research indicates that muscle fatigue can be identified by analyzing features of the sEMG signal. Fuentes del Toro et al. used wavelet filtering and evaluated RMS (Root Mean Square), MAV (Mean Absolute Value), and MNF (Mean Frequency). Showing that RMS and MAV increased while MNF decreased as fatigue developed [3].

This project aims to evaluate an sEMG measurement system that amplifies, filters, and records muscle activity. The recorded data is analyzed over time using RMS, MAV, and MNF to investigate whether a consistent fatigue related pattern can be identified.

Design

This project presents a wearable sEMG measurement system consisting of three electrodes connected to an elastic band with a signal processing device as shown in figure 2. The electrodes detect the electric signals generated by the muscle, which are then amplified and filtered using an analog circuit. An Arduino converts the conditioned analog signal to a digital signal by using its built-in ADC, which is controlled by C++ code. The acquired data can be transferred via USB to a computer, where a python program calculates RMS, MAV and MNF, and plots the results for analysis of muscle activation and fatigue. The Arduino is also intended to transmit the data wirelessly via Bluetooth to a smartphone, allowing the processed results to be presented through a mobile application for the user [figure 1].

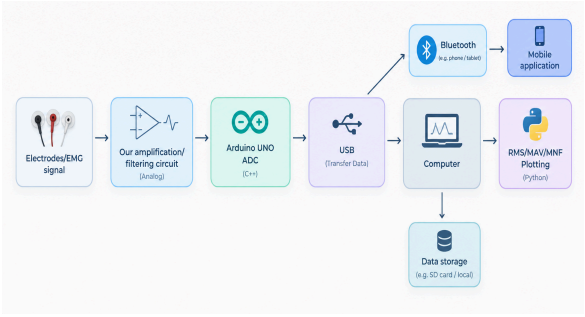


Figure 1: System overview of the signal flow from sEMG electrodes to signal processing, data transfer and result visualization (Image generated by AI).



Figure 2: Conceptual design of the wearable sEMG measurement system (Image generated by AI).

Evaluation

We first evaluate the sEMG measurement system using a function generator and an oscilloscope to verify the amplification and filtering. Afterwards, the output of the developed circuit is sent to a PicoScope oscilloscope and records the sEMG signal from the muscle. We attach three electrodes to the muscle and observe the signal while the muscle gradually contracts. Figure 3 shows one of the recorded signals. Where the first part of the measurement shows the muscle in a relaxed state followed by an increased muscle activation.

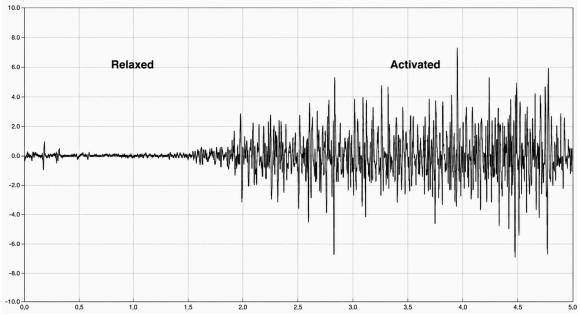


Figure 3: sEMG signal recorded during muscle contraction. Showing the increase in signal amplitude as muscle activation increases over time.

As shown in figure 3, the signal remains close to the baseline when the muscle is relaxed. As the muscle contraction increases, the amplitude of the signal also increases. This indicates that the system is capable of detecting alterations in the electrical activity of the muscle. However, the increase in amplitude is not sufficient enough to determine whether the muscle is fatigued. Further signal processing and analysis is required to identify any fatigue related changes in the recorded signal.

The next stage of the evaluation will therefore focus on calculating RMS, MAV and MNF from the recorded data. Based on the previous research provided in the introduction, an increase in RMS and MAV with a decrease in MNF is expected as muscle fatigue develops. Comparing these parameters at different periods of the contraction could make it possible to identify patterns and distinguish between a fatigued and non fatigued muscle.

Conclusion

The problem of measuring muscle fatigue is addressed by developing an sEMG measurement system that amplifies, filters, records and analyzes muscle activity using RMS, MAV and MNF. Initial results indicate that the system can detect changes in muscle activation. However, further analysis of RMS, MAV and MNF is required to determine whether consistent patterns can be used to distinguish fatigued from non fatigued muscle. The system will be validated by performing controlled muscle fatigue experiments and comparing the calculated RMS, MAV and MNF values between different exercise repetition to determine whether the expected fatigue related patterns can be consistently observed.

References

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Doi: https://doi.org/10.3390/s19143204